

CSE211s Introduction to Embedded Systems

Spring 2026

Lecture Topics

- Embedded Systems & Applications
- General-Purpose vs Embedded Computers
- Embedded Systems
 - Fields of Application
 - Design Considerations
- Basic Components of a Digital Computer
 - CPU – Memory – IO
 - Buses: Address – Data – Control
- Embedded Systems Development Tools
 - Software Tools
 - Hardware Toole

Embedded Systems

*Implanted inside
something*



► An embedded system:

- ↳ It is a digital computing device (CPU + Memory + IO)
 - ↳ embedded within a system
 - ↳ to provide 1) *enhanced functionality*
2) *ease of use*.
- ↳ It executes software encoded in its flash memory (*firmware*) capable of
 - ↳ carrying complex tasks that are almost impossible by discrete logic circuits.

Should users interact with the system, this interaction should come in a natural way.

Real Embedded System Example

Digital Camera

User point of view

A camera

A digital camera

Traditionally ...

You press a button; you take a photo, but before that you need to manually:

1. Adjust the focus according to the distance
2. Adjust the speed of the shutter according to light intensity and relative speed
3. Decide on using a flashlight according to light intensity.

Ease of use:

You press a button; you take a photo.

1. Focus is adjusted automatically using infrared source/sensor, and a motor to position the lens.
2. Speed of the shutter is adjusted according to the output of a light intensity detector.
3. Flashlight is on automatically if the the light intensity detector is below a minimum threshold.

Enhanced functionality

without exposing the user to the complexities of a computer:

- 1.Reduce blur caused by camera shake.
 - 2.Image review / zoom / edit
 - 3.Multiple exposures to create images with a wider range.
- •
•

Embedded Systems

Fields of Applications

- **General Computing**

Applications like desktop computing, but in an embedded package

- Video games, wearable computers, automatic tellers
- Tablets, Phablets

- **Control Systems**

Closed loop feedback control of real time system

- Vehicle engines, chemical processes, nuclear power, flight control, building access control, smart homes

- **Signal Processing**

Computations involving large data streams

- Video, Audio, Radar, and Sonar

- **Communication & Networking**

Switching and information transmission

- Telephone system, Internet
- Wireless everything, internet of things (IoT)

Embedded Systems All Around

Tablets



Wearable
Computers



Traffic
Control



Biomedical

Smart Phones



Video Games



Networks



Robotics



Consumer Electronics



Home
Appliances



Transportation

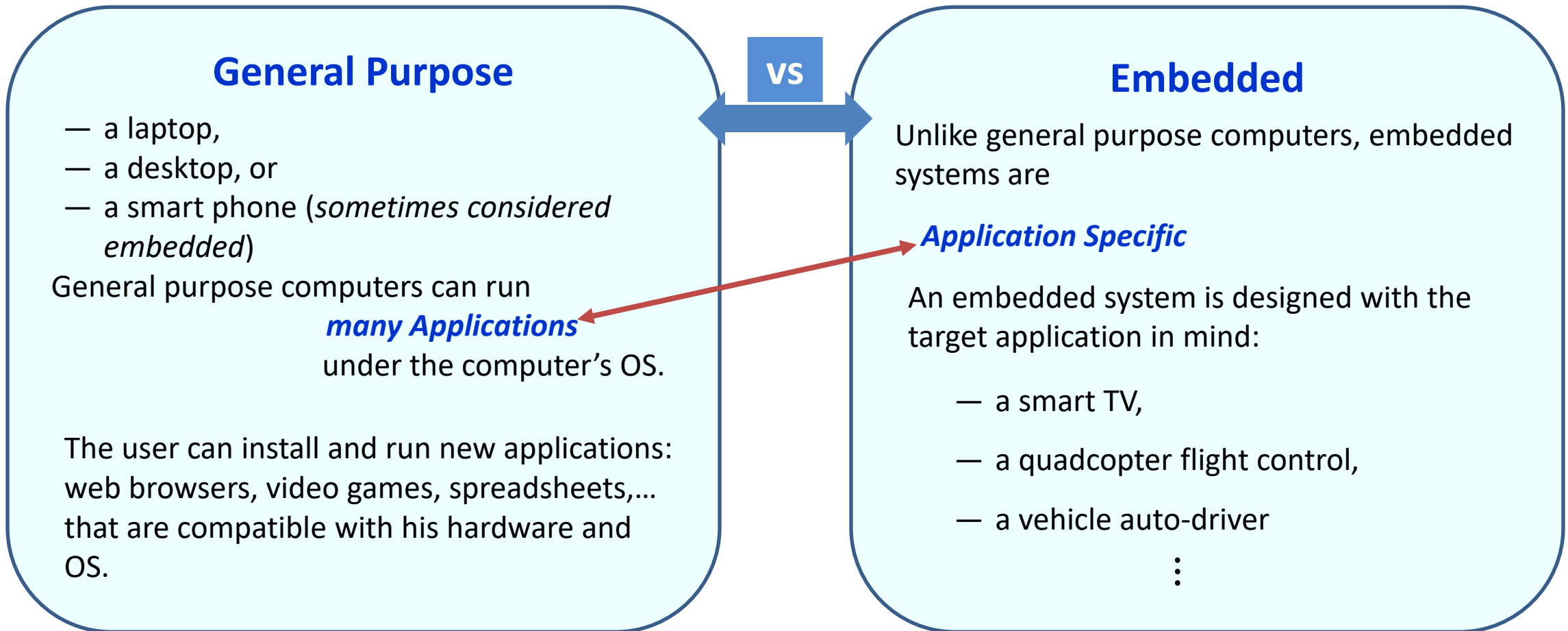


Telephony



Automotive

Embedded vs General Purpose Computers



- In an embedded system, hardware and software are **codesigned** to optimize performance.
- For example, an Xbox console runs a video game more efficiently than a laptop under Windows because it is designed specially for graphics and video processing.

Embedded Systems Interfaces

Embedded systems typically:

- Input data from the user, if required, through suitable interfaces, and from the surrounding through suitable sensors
- Process the data and take decisions
- Output control actions to control the system through actuators.

Hence an embedded system usually has the following interfaces:

1. User interface:

To allow the user/operator to *set* and *get* the state of the system

2. Real World interface:

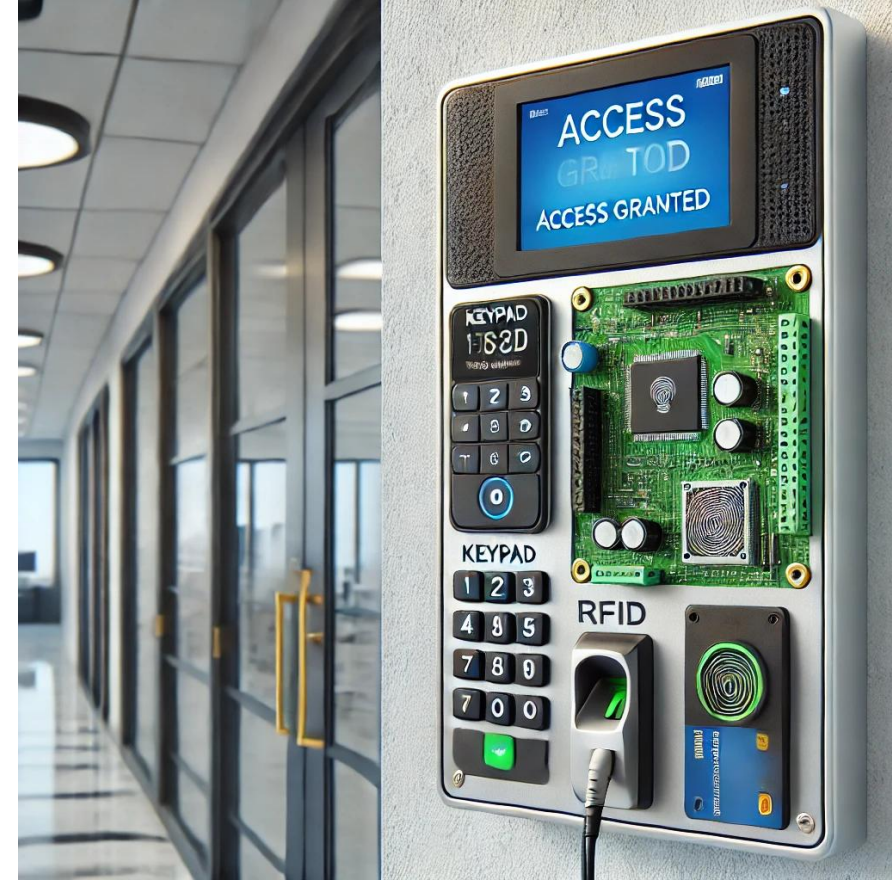
- To allow the embedded computer to read/measure the *state of the 'world'* through sensors (digital or analog)
- To influence *the state of the 'world'* through actuators (digital or analog)

3. Communication interface (optional):

Connect to other devices, embedded devices or the Internet

Real Embedded System Example

Access Control (Building/Premises/Room)



Real Embedded System Example

Access Control (Building/Premises/Room)

Example workflow:

1. Input:

A user presents his credentials such as RFID card to the RFID Reader

2. Processing:

The system verifies the credential against a database.

3. Output:

If valid, it activates the Electric Lock and displays "Access Granted" on the LCD.

If invalid, it triggers the buzzer (**Output**) and denies access.

Real Embedded System Example

Access Control (Building/Premises/Room)

Inputs (Sensors & User Interaction Devices)

These components collect data to determine access permission.

1. User presenting device options:

2. Keypad – Users enter a PIN code for authentication.
3. RFID/NFC Reader – Reads access cards or key fobs.
4. Biometric Scanner – Fingerprint, facial camera, or iris scanner.
5. Magnetic Stripe Reader – Reads traditional access cards.
6. Bluetooth Receiver – Communicates with smartphones for keyless entry.
7. Motion Sensor – Detects presence near the access point.
8. Microphone & Speaker – For voice recognition or intercom communication.

2. Gate status sensor:

1. Door/gate Contact Sensor – Checks if the door/gate is open or closed.
2. Push Button (Exit Button) – Allows users to exit without authentication.

3. Security logs:

1. Camera CCTV (Closed-Circuit Television) Integration – Captures images or videos for security logs.

Real Embedded System Example

Access Control (Building/Premises/Room)

Outputs (Control & Notification Devices)

These components execute actions based on input validation.

1. Options to Control physical access:

2. Electric Lock (Solenoid Lock or Magnetic Lock)
3. Automatic gates or barriers control (Servo/Motor Driver)

2. Alarms and indicators:

1. Alarms to sound alerts for unauthorized access attempts or gates (Relay Output).
2. System status indicators (LEDs, Green: Access Granted, Red: Denied).
3. Displays messages – (LCD/TFT Display, "Access Granted" or "Invalid Entry").
4. Voice prompts or intercom functionality (Speaker System)

3. Connectivity:

1. Network Interface (Wi-Fi/Ethernet/GSM) – Sends access logs to a central server.
2. Bluetooth Transmitter – Sends notifications to authorized devices.
3. SMS/Email Notification – Alerts security or building managers on suspicious activity.

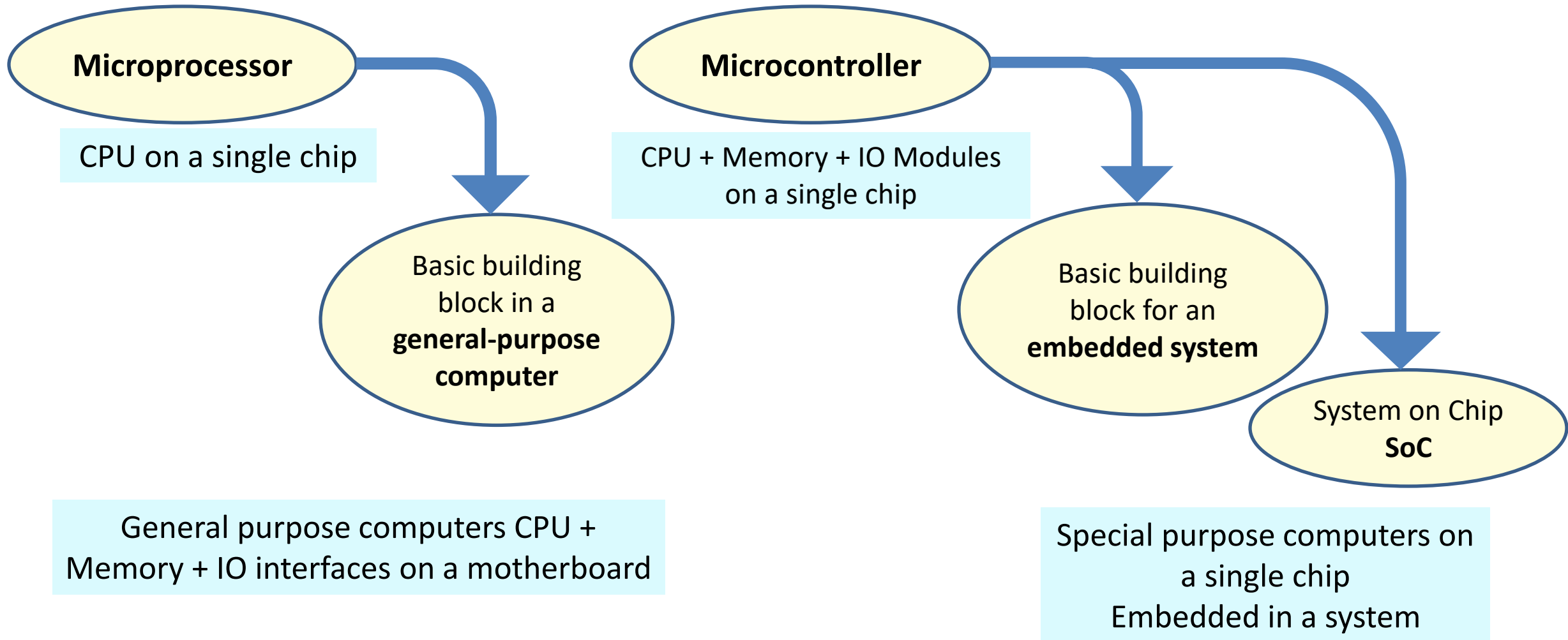
Options for Building Embedded Systems

	Function Complexity	Design Cost	Upgrades & Bug Fixes	System Speed
1. Dedicated hardware: • Discrete Logic • ASIC • Programmable logic – FPGA, PLD	Low to moderate	• Low • High • Low	Hard	Fast
2. Software running on generic hardware: • Microcontrollers	Complex	Low to moderate	Easy	Moderate

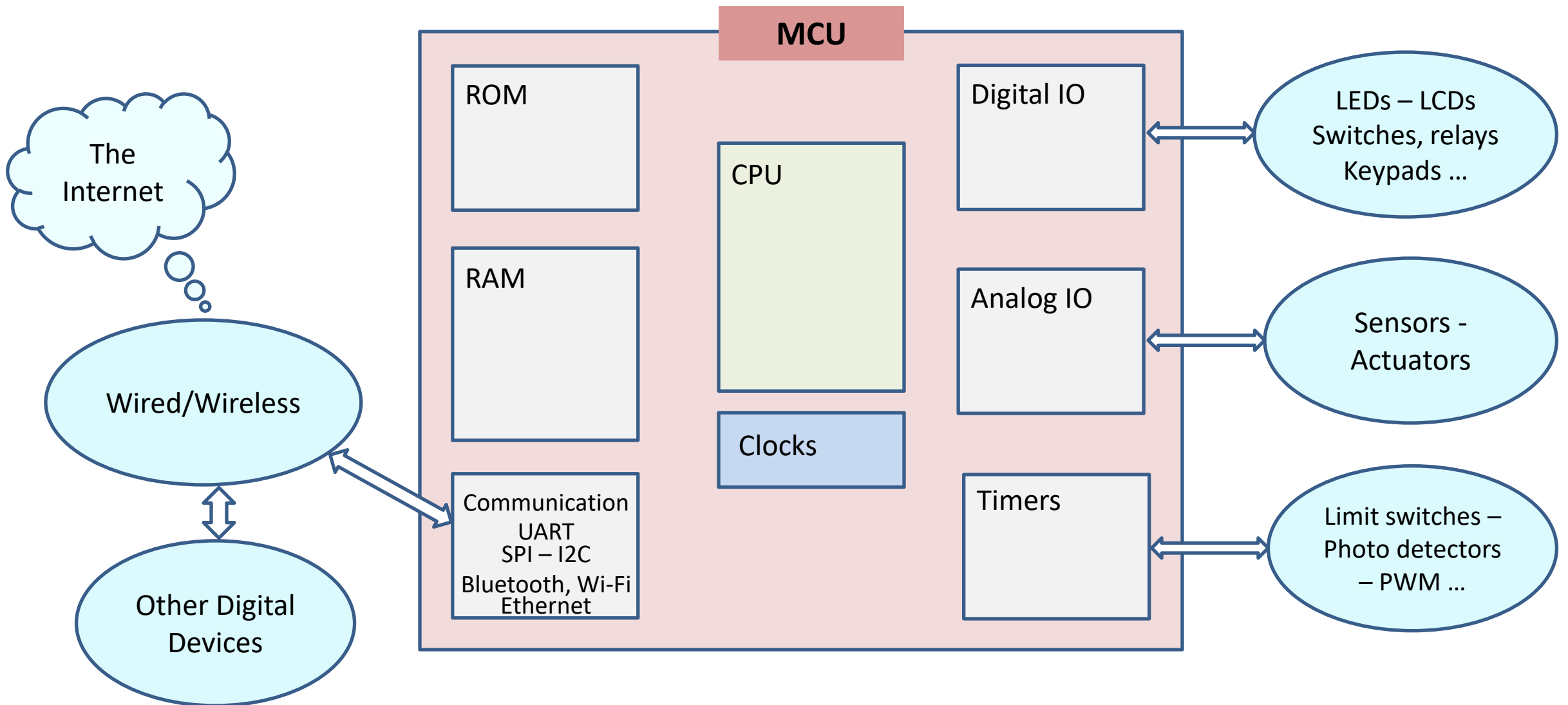
Building Embedded Systems Using Microcontrollers

- In most embedded systems, MCUs are the best solution as they offer:
 - Low development and manufacturing cost
 - Easy porting and updating
 - Light footprint
 - Relatively low power consumption
 - Satisfactory performance for low-end products

Terminology



Typical Microcontroller Components



Typical Embedded System Considerations

- **Small Size, Low Weight**

- For handheld electronics

- **Low Power**

- Battery operated
- Limited cooling may limit power

- **Low Cost**

- \$.05 adds up over 1,000,000 units

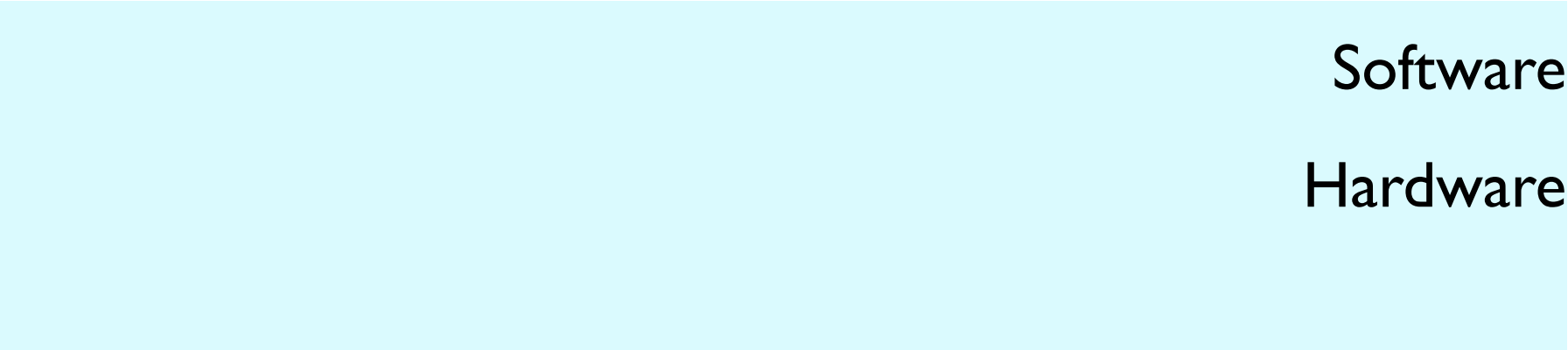
- **Working in Harsh environment**

- Heat, vibration, shock
- Power fluctuations, RF interference, lightning
- Water, corrosion, physical abuse

- **Safety critical operation**

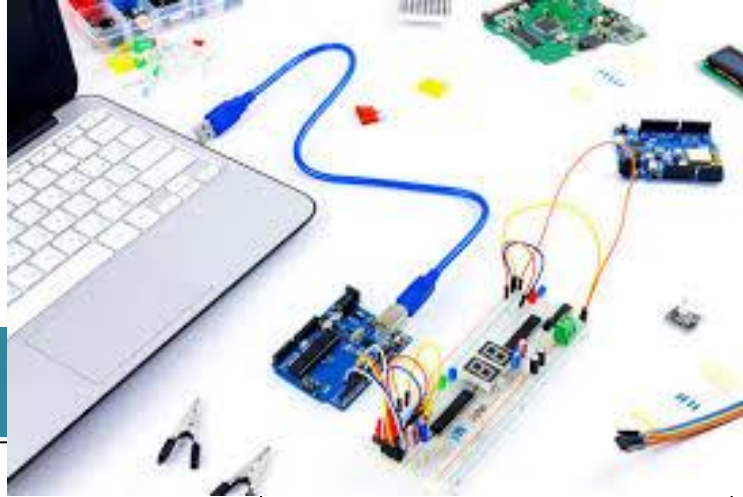
- Must function correctly
- Must not function incorrectly

Embedded Systems Development Tools



Software
Hardware

MCU Development Tools



Software

On your PC:

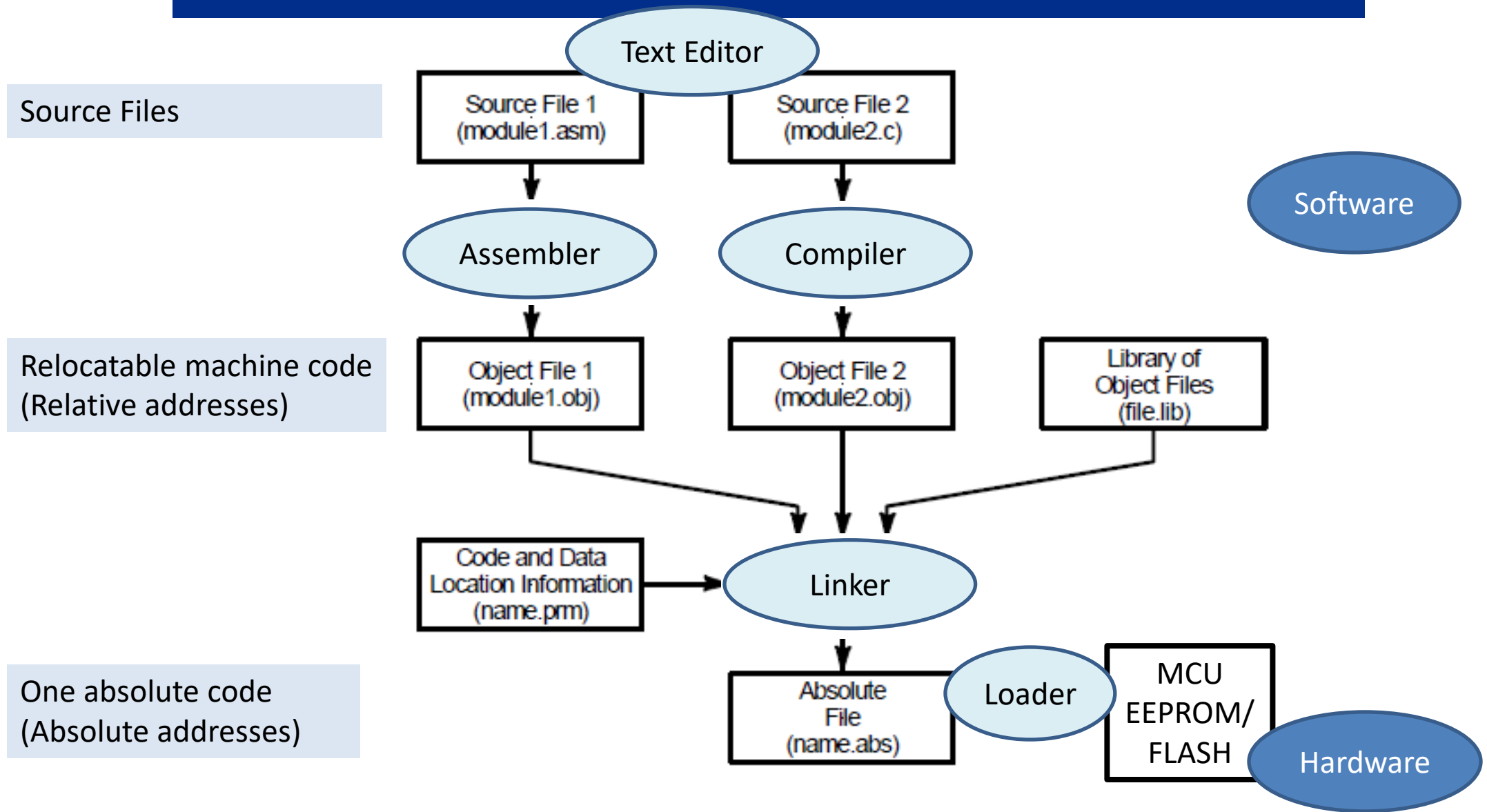
- Text editor
- Cross assembler
- Cross compiler
- Simulator
- Source-level debugger

Hardware

- Demo boards
- Development boards
- Evaluation boards
- Cables (USB / RS232)
- Pods

Integrated development
environment (IDE)

Embedded Software Development



Embedded Software Development Tools

Text Editor:

It is the primary interface where developers write, edit, and manage source code. (Assembly, C, ...)

Assembler:

Translates assembly language code into machine code (binary instructions) that the microcontroller can execute.

Compiler:

Translates high-level programming languages into assembly language or machine code (binary instructions) that the target microcontroller can execute.

Linker:

Converts multiple object files into a single executable file that can run on the target hardware.

Loader:

Loads executable code into memory and transfers control to the program's entry point.

Extra

Software Development Tools

On your host PC: (possibly on the cloud, e.g. *Keil Studio Cloud*)

- A **text editor** to enter and edit C and/or Assembly code.
- A **cross compiler/assembler** to compile/assemble the source program.
- A **simulator** to run the application program without having the actual hardware.
- A **terminal program** to communicate with the hardware board if exists.
- A source-level **debugger** to set breakpoints in the program, trace program execution, watch variable values ...

The IDE combines all the above tools in one package so the user can perform all software developing activities in one environment.

Hardware Development Tools

A development board:

- A MCU
- A USB, or serial port, with cable to a host PC where the user develops the application program
- An on-board D-Bug monitor, or Serial monitor program, in the MCU flash memory to communicate with the PC through the USB.

The D-Bug / Serial monitor program allows the user to:

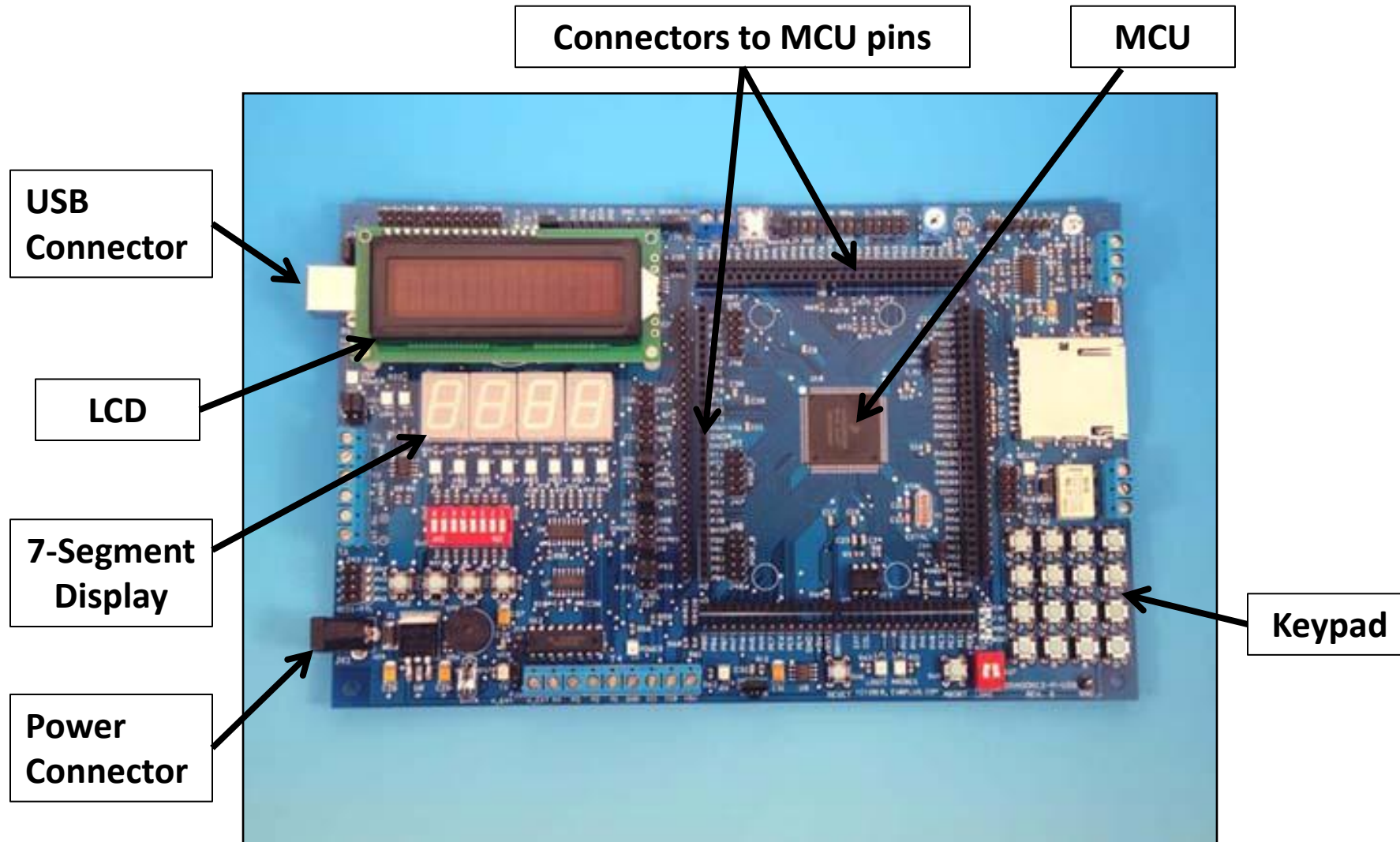
- Download programs onto the EVB MCU memory for execution
 - Display/Set register and memory location contents
 - Set program breakpoints
 - Trace instruction execution
- Many peripheral connectors to MCU pins such as digital IO, analog IO, timers, serial communication ...

Some development boards may have:

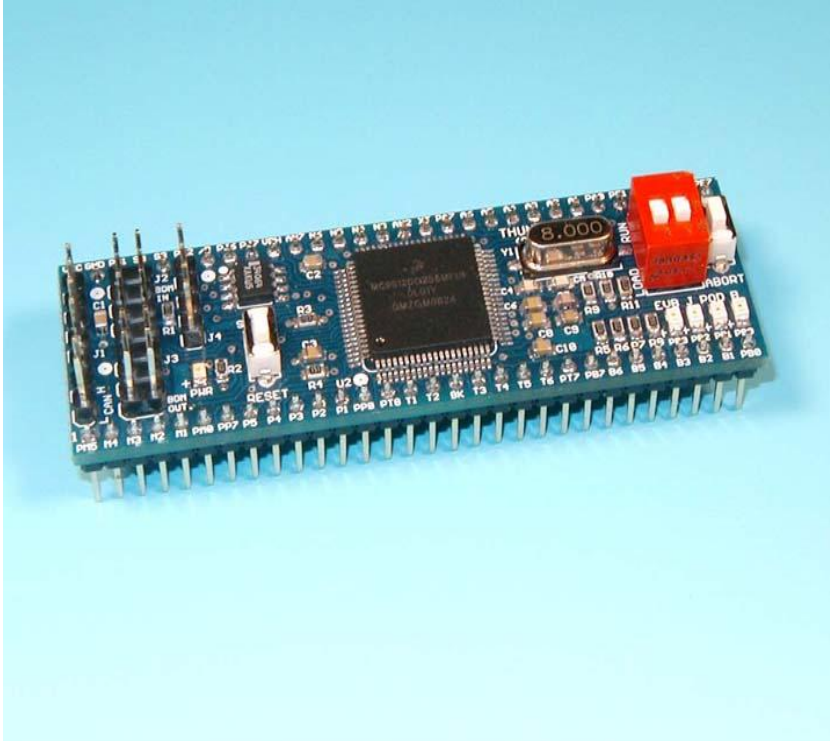
- Input capabilities through a keypad, switches, push buttons, touch screen ...
- Output through LEDs, 7-segment displays, LCDs ...

MCU Evaluation Boards (EVBs)

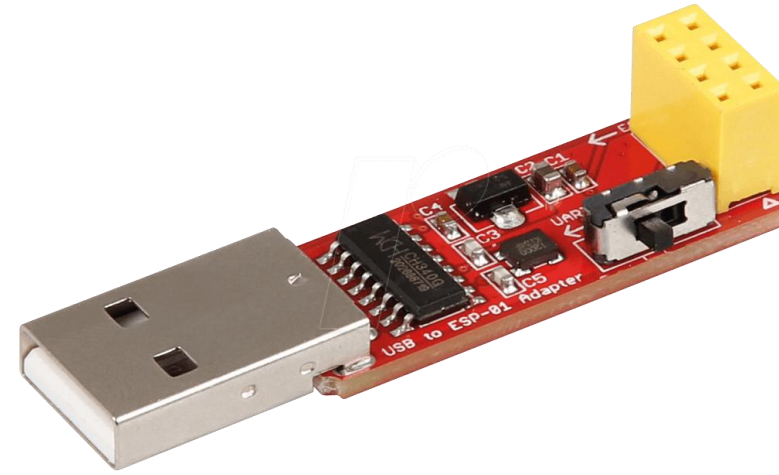
Example: The Dragon12-Plus-USB



Cheaper HWD Development



The ThunderBird12 DIP module
No power supply or USB included



DEBO ESP8266 USB - Development boards -
ESP8266 USB programmer
USB interface to host PC provides power

Our Choice: STM32 Nucleo-64 Development Board

with STM32F401RE MCU

➤ Common features:

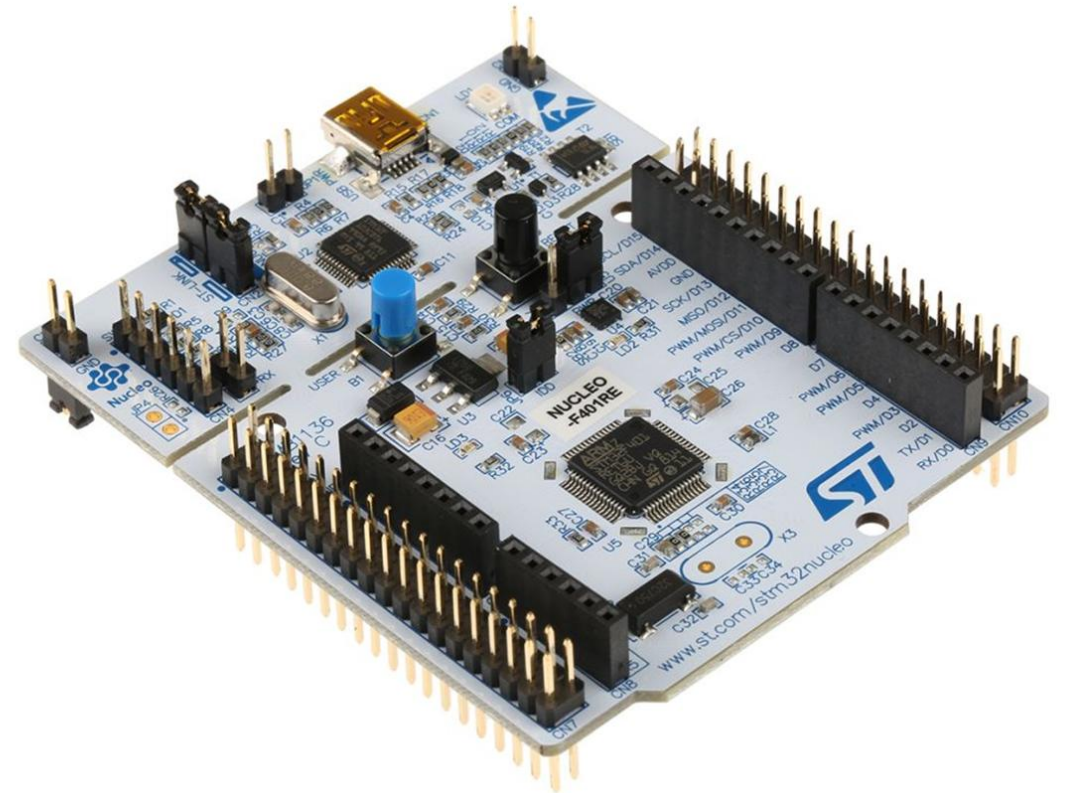
- STM32 microcontroller
- 1 user LED shared with ARDUINO®
- 1 user and 1 reset push-buttons

➤ Board connectors:

- ARDUINO® Uno V3 expansion connector
- ST morpho extension pin headers for full access to all STM32 I/Os
- Flexible power-supply options: ST-LINK USB V_{BUS} or external sources
- On-board ST-LINK debugger/programmer with USB re-enumeration capability:
- mass storage, Virtual COM port, and debug port
- Support of a wide choice of Integrated Development Environments (IDEs)
- including IAR Embedded Workbench®, MDK-ARM, and STM32CubeIDE
- External SMPS experimentation dedicated connector
- USB Type-C®, Micro-B, or Mini-B connector for the ST-LINK
- USB Type-C® user connector
- MIPI® debug connector

➤ Board-specific features

- External SMPS to generate V_{core} logic supply
- 24 MHz or 48 MHz HSE
- User USB Device full speed, or USB SNK/UFP full speed
- Cryptography
- Board connectors:



STM32F401RE MCU Features

- Core: ARM® 32-bit Cortex®-M4 CPU with FPU, clock frequency up to 84 MHz, and DSP instructions
- Memories:
 - up to 512 Kbytes of Flash memory
 - up to 96 Kbytes of SRAM
- Voltages: 1.7 V to 3.6 V application supply and I/Os
- Analog Interfaces: One 12-bit, 2.4 MSPS A/D converter: up to 16 channels
- General-purpose DMA
- Up to 11 timers: up to six 16-bit, two 32-bit timers up to 84 MHz and a SysTick timer

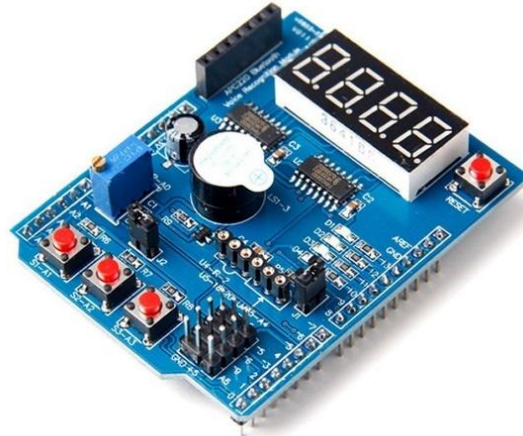
- Debug mode
- Up to 81 I/O ports with interrupt capability, 5 V-tolerant
- Communication interfaces
 - I²C interfaces
 - USARTs
 - SPI Interfaces
 - SDIO interface
 - USB 2.0

STM32F401RE MCU

Family	ARM Cortex-M4
Vendor	ST Microelectronics
RAM	96KiB
Flash	512KiB
Frequency	up to 84MHz
FPU	yes
Timers	11 (2x watchdog, 1 SysTick, 6x 16-bit, 2x 32-bit [TIM2])
ADCs	1x 12-bit (16 channels)
UARTs	3
SPIs	3
I2Cs	3
RTC	1
USB	1
Vcc	1.7V - 3.6V
Datasheet	Datasheet
Reference Manual	Reference Manual
Programming Manual	Programming Manual
Board Manual	Board Manual

Arduino Shield Samples

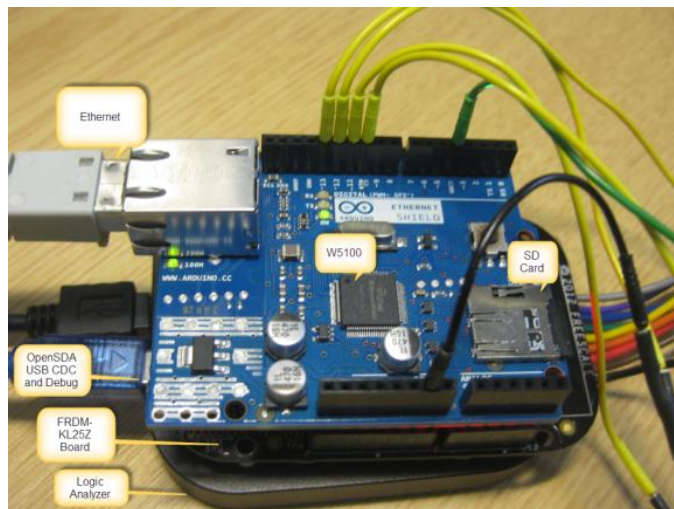
Multifunction
Arduino Shield



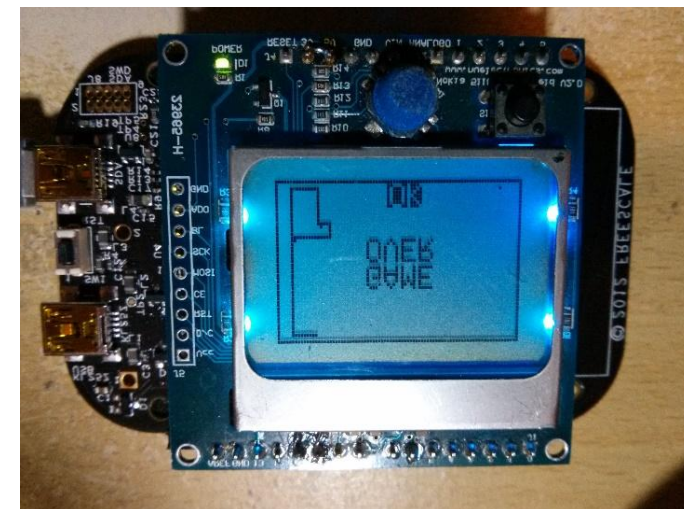
Stepper Motor
Arduino Shield



Ethernet
Arduino
Shield



LCD and joystick
Arduino shield





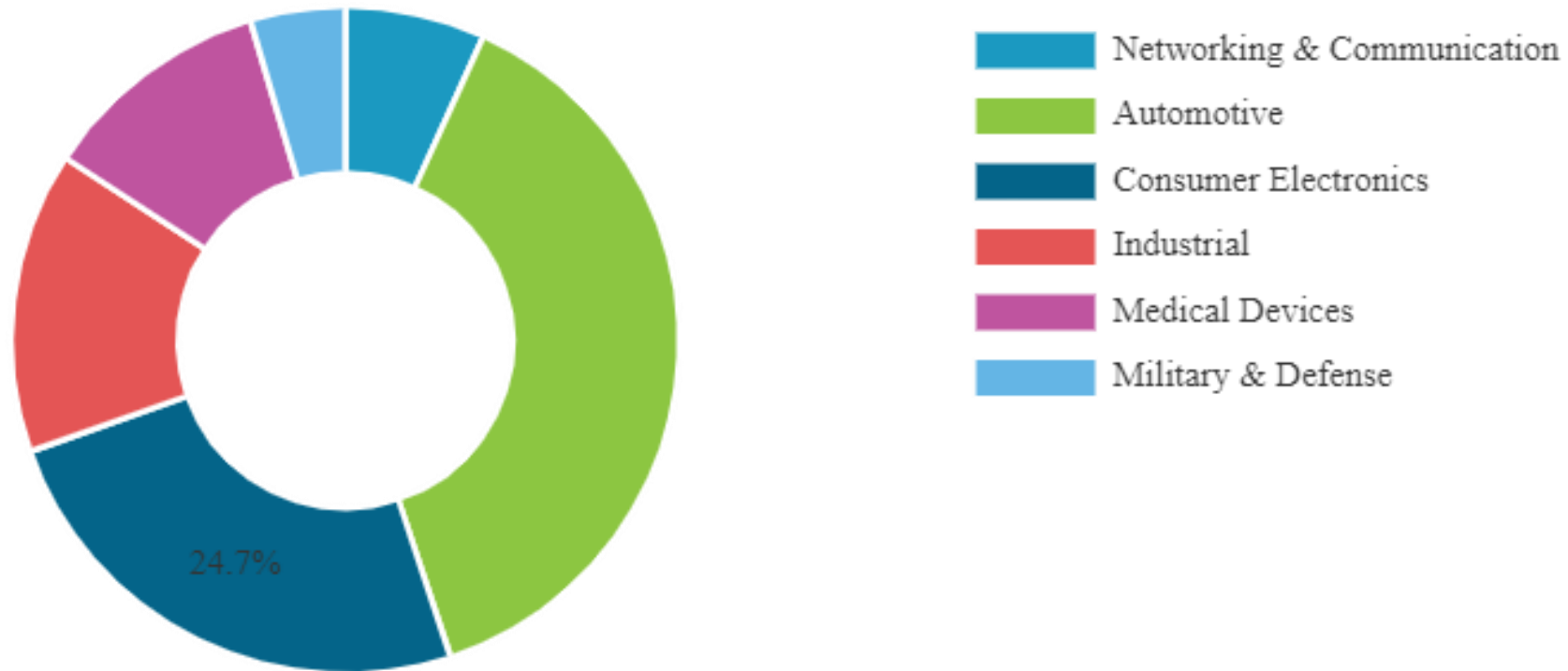
Course's IDE: Arm® Keil® Studio Cloud

Arm® Keil® Studio Cloud is a free, browser-based IDE for the evaluation and development of embedded, IoT, and Machine Learning software for **Cortex®-M** devices.

User can edit C/C++ projects from any computer, share them with colleagues and export them for desktop usage in the Keil µVision® software development platform. User can compile projects and run them directly on supported development boards, and debug from supported browsers without the need to install any software.

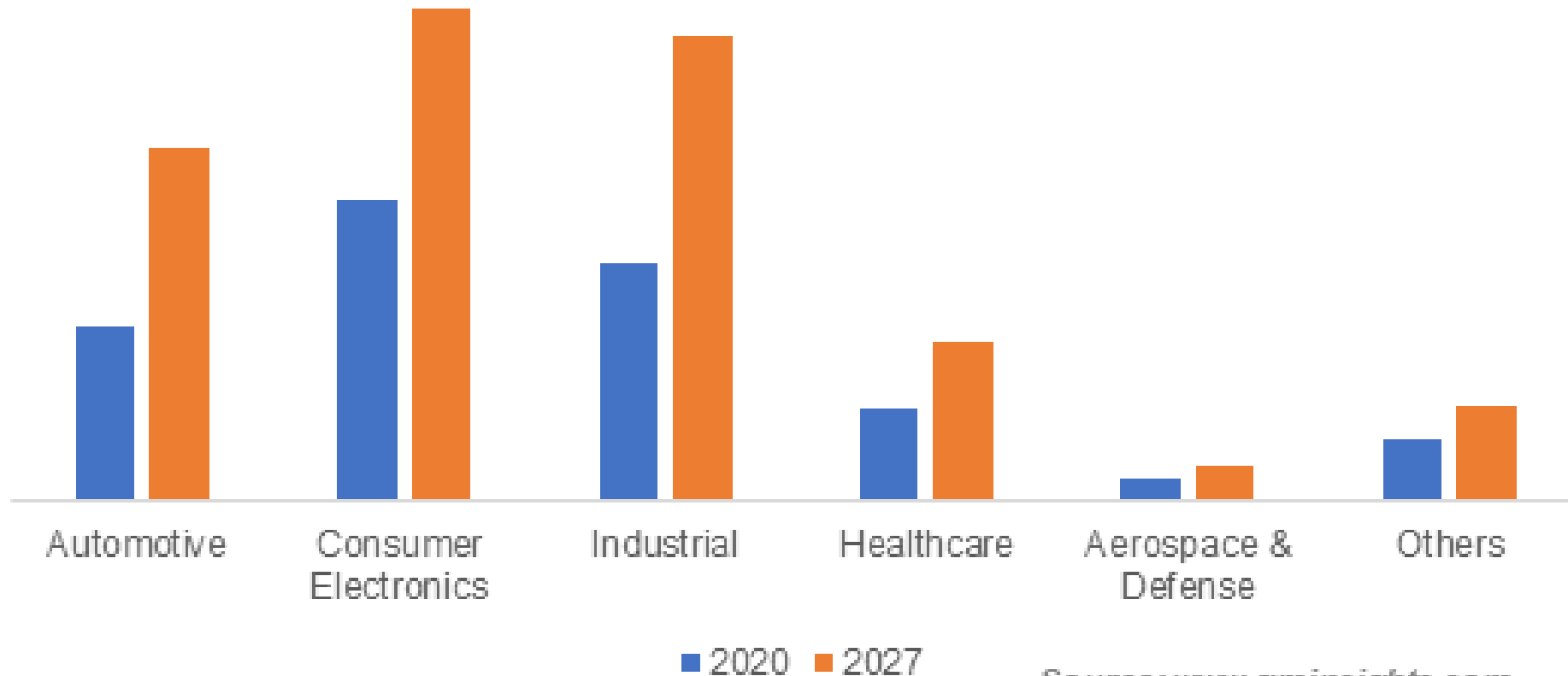
Microcontrollers Market Share by Application

Global Microcontroller Market Share, By Application, 2020



Microcontrollers Market Forecast by Application

U.S. Microcontrollers Market Size, By Type, 2020 & 2027 (USD Million)



Source: www.gminsights.com

Microcontrollers Market Share by Manufacturer

Leading MCU Suppliers (\$M)

2021 Rank	Company	Headquarters	2020	2021	21/20 % Chg	2021 Marketshare
1	NXP	Europe	2,980	3,795	27%	18.8%
2	Microchip	U.S.	2,872	3,584	25%	17.8%
3	Renesas	Japan	2,748	3,420	24%	17.0%
4	ST	Europe	2,506	3,374	35%	16.7%
5	Infineon	Europe	1,953	2,378	22%	11.8%

Source: Company reports, IC Insights

STMicroelectronics commonly referred to as **ST** or **STMicro**) is the largest European semiconductor manufacturing and design company.

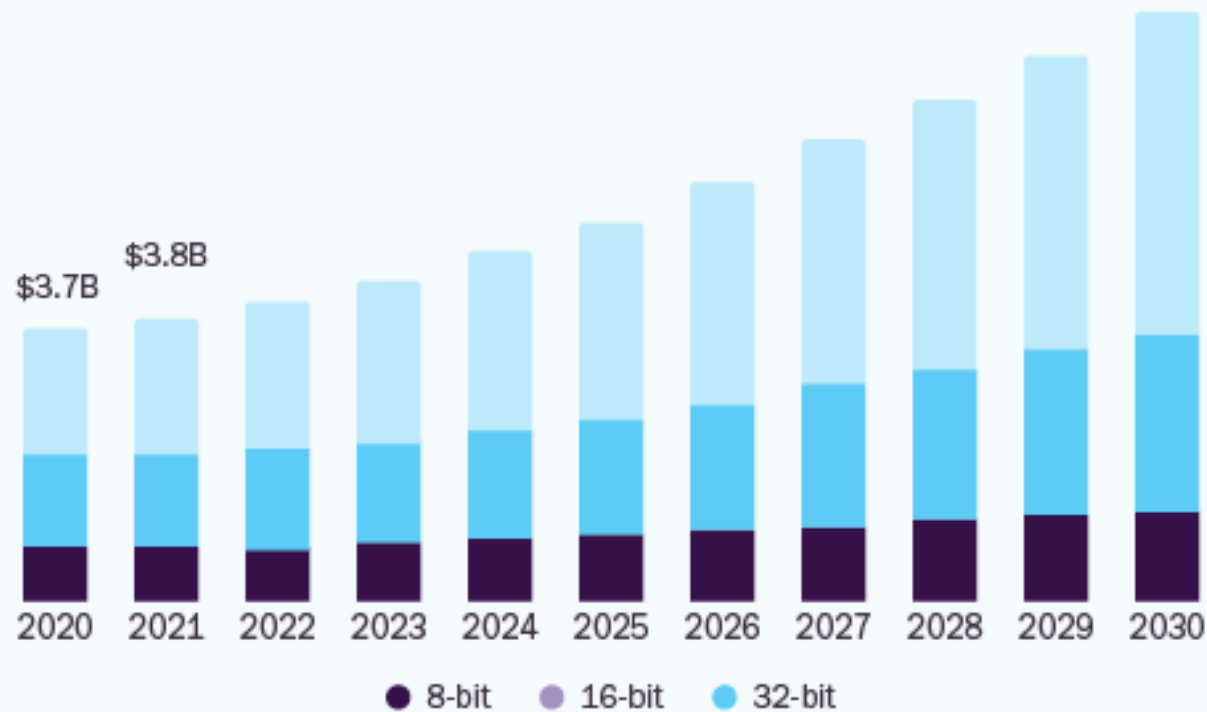
The company resulted from the merger of two government-owned semiconductor companies in 1987: “Thomson Semiconducteurs” of France and “SGS Microelettronica” of Italy.

ST opened a center in **Maadi Technology Park** in Egypt for electronics design and software development.

Microcontrollers Market Share by Size

U.S. Microcontroller Market

size, by product, 2020 - 2030 (USD Billion)

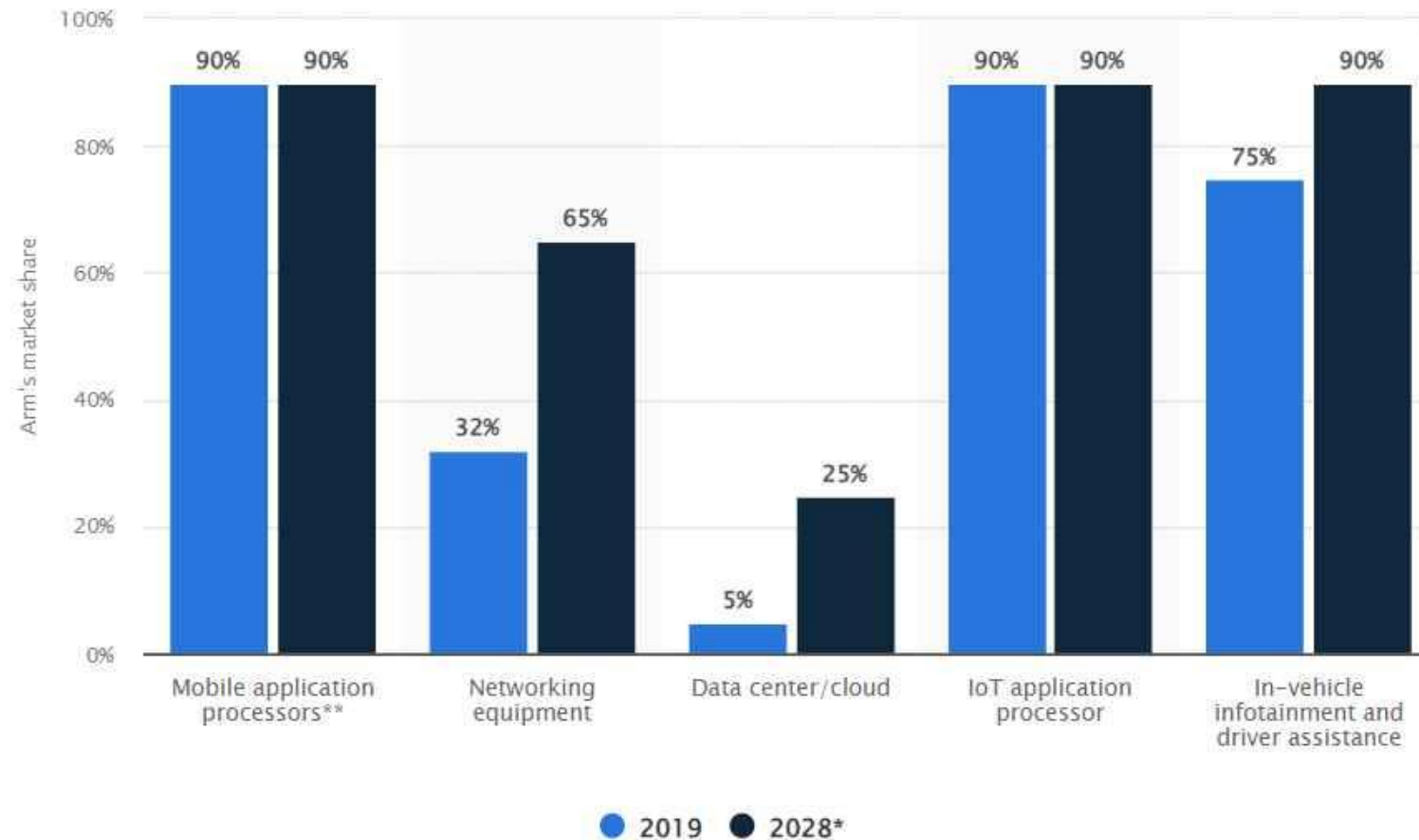


9.0%

U.S. Market CAGR,
2022 - 2030

Source:
www.grandviewresearch.com

Arm's market share and targets across key technology markets



Where Could You End Up?

- **Automotive systems**
 - ✓ perhaps designing and developing “drive-by-wire” systems
 - ✓ self-driving vehicles
- **Telecommunications**
- **Consumer electronics**
 - ✓ cellular phones, MP3 devices, smart TVs, tablets
 - ✓ Home appliances
 - ✓ Internet appliances (your washer will be on the internet more than you are!)
- **Defense and weapon systems**
- **Process control**
 - ✓ chemical refinement
 - ✓ food industry
- **Automated manufacturing**
 - ✓ Supervisory Control and Data Acquisition (SCADA)
- **Space applications**
 - ✓ Satellite communications